FINAL YEAR CAPSTONE PROJECT • 2025–2026

SEO Audit (WebIntel)

An AI-Powered Website Intelligence, Deterministic SEO Auditing, Traffic Estimation & Competitor Benchmarking Platform

PRESENTER

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FULL-STACK CORE

Next.js 16 + React 19

TypeScript • Tailwind CSS v4

Server Components & Edge Actions

CLOUD INFRASTRUCTURE

Google Cloud Run

Cloud SQL PostgreSQL 16

Serverless Container Autoscaling

Live Production URL: `https://seoaudit-93977081206.us-central1.run.app`

PRODUCTION LIVE

< 100ms	100% Deterministic	50+ Signatures	FastMCP

PROBLEM STATEMENT & INDUSTRY GAPS

The Triple Breakdown in Modern Web Auditing

Current SEO software and diagnostic ecosystems suffer from closed access, opaque algorithms, and massive server resource exhaustion.

FLAW 01

Closed Paywalls

Standard web inspection and technical diagnostics are locked behind closed commercial paywalls, denying developers, students, and startups access to their own website telemetry.

Closed Ecosystems

FLAW 02

Opaque & Fabricated Math

Proprietary ratings hallucinate exact visitor numbers and domain authority without disclosing confidence bounds, error margins, or underlying formulas.

Black Box Ratings

FLAW 03

Server Resource Bloat

Traditional crawlers launch headless Chromium (Puppeteer/Playwright) consuming 1.5–2 GB RAM per crawl, causing severe latency and frequent server memory crashes.

ENGINEERING PHILOSOPHY & INTEGRITY

Zero-Hallucination Architectural Standards

A high-performance SaaS platform delivering honest, verifiable website diagnostics across five analytical pillars.

Core System Guarantees

- **Sub-Second Crawl Latency (<100ms):** Lightweight Cheerio DOM parser extracts headers and tags without browser overhead.
- **Deterministic 0–100 SEO Score:** Transparent mathematical formula divided into 4 equal 25-point categories.
- **50+ Signature Tech Stack Detector:** Fingerprints CMSs, frameworks, CDNs, and server runtimes in milliseconds.
- **Core Web Vitals Telemetry:** Google PageSpeed Insights REST API measures LCP, CLS, INP, and TTFB.

Zero-Hallucination AI Philosophy

AI is strictly an analytical interpreter, never the source of truth for numbers.

- ✓ Deterministic code computes all raw metrics, status codes, and audit checks.
- ✓ Google Gemini AI interprets verified findings to synthesize executive summaries & copyable developer code fixes.
- ✓ Automated Heuristic Rule Synthesizer activates if external APIs time out.

FULL-STACK ARCHITECTURE & MICROSERVICES

6-Layer Modular Pipeline & Data Flow

Microservice containerization deployed on Google Cloud Run with Google Cloud SQL PostgreSQL 16.

6-Layer Pipeline

Layer 1: Next.js 16 App Router + React 19 UI	SSR / Client
Layer 2: SSRF Security Shield & Pre-Flight DNS	Zero Trust
Layer 3: Cheerio HTML DOM & Header Parser	< 100ms
Layer 4: Deterministic Scorer & PageSpeed API	0-100 Formula
Layer 5: Google Gemini AI Diagnostic Synthesizer	Code Fixes
Layer 6: Google Cloud SQL PostgreSQL 16	UUID /report/[id]

```
// Complete Request Flow

Target URL Submission
  |
  ▼
[SSRF DNS Guard] ──▶ Drops 127.0.0.1 / 169.254.x
  |
  ▼
[Cheerio Engine] ──▶ Sub-100ms DOM Parse
  |
  ├─▶ 0-100 Category Scorer (4 Quadrants)
  ├─▶ 50+ Tech Detector (Signatures)
```

MATHEMATICAL FORMULATION & SCORING ENGINE

The 0–100 Deterministic Scoring Model

A transparent, 4-quadrant mathematical formulation evaluating 25 distinct technical factors with zero bias.

QUADRANT 1 • 25 PTS

Metadata & Headers

- Title tag (10–60 chars)
- Meta Description (50–160 chars)
- Viewport & Mobile Meta
- OpenGraph & Twitter Cards
- Charset & Favicon declared

QUADRANT 2 • 25 PTS

Content & Hierarchy

- Single H1 tag present
- H2–H6 logical hierarchy
- Image alt attribute coverage
- Word count threshold (>300w)
- Content-to-HTML ratio

QUADRANT 3 • 25 PTS

Indexability

- Canonical link matches URL
- Robots.txt reachable
- XML Sitemap discovered
- Noindex tag absence
- Internal vs External link ratio

ENTERPRISE SECURITY & NETWORK HARDENING

Enterprise SSRF Defense & Safe Crawling

Server-Side Request Forgery mitigation implemented at the pre-flight network resolution level.

The SSRF Threat Model

Public URL scrapers are highly vulnerable to attackers submitting internal addresses to exfiltrate database credentials or cloud IAM tokens.

- **RFC 1918 Private Subnets:** 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
- **Loopback & Localhost:** 127.0.0.1, ::1, 0.0.0.0
- **Cloud Metadata APIs:** 169.254.169.254, metadata.google.internal

```
// SRC/LIB/SECURITY/SSRF.TS
const ips = await dns.resolve4(parsed.hostname);

for (const ip of ips) {
  if (isPrivateOrReservedIp(ip)) {
    throw new SecurityException(
      "SSRF_BLOCKED: Resolved internal IP"
    );
  }
}

// Timeout: 8000ms socket abortion
// Payload Limit: 5 MB stream cutoff
```

Enforces pre-flight DNS lookups to defeat DNS Rebinding attacks before network connection.

TRAFFIC ESTIMATION & STATISTICAL MODELING

Honest Traffic & Organic Authority Estimation

Modeling domain audience reach and search visibility using transparent CTR decay curves without fabricated numbers.

The Mathematical Traffic Formulation

We compute organic traffic estimates using empirical SERP position click-through curves:

```
Estimated Traffic = Σ (Keyword Volume × Position CTR)
• Pos 1: ~31.7% CTR  |  Pos 2: ~24.7% CTR
• Pos 3: ~18.6% CTR  |  Pos 4-10: Logarithmic decay
```

Domain authority is evaluated via backlink diversity, protocol security, and content indexability signals.

Statistical Confidence Bands

High Confidence Level
Direct search console feeds or verified indexing telemetry with <10% variance.

Medium Confidence Level
Statistical crawl extrapolation factoring backlink volume and indexed pages.

Estimated Bounds Interval
Displays lower and upper standard deviations instead of false precision.

COMPARATIVE INTELLIGENCE & FINGERPRINTING

Side-by-Side Benchmarking & 50+ Tech Profiler

Sub-100ms technology stack detection and head-to-head metric battle between target and rival domains.

Head-to-Head Comparative Engine

Compares two target URLs concurrently with deterministic differential analysis:

SEO Health Score Delta	**Target (+12 pts higher)**
Time to First Byte (TTFB)	**Target (82ms vs 310ms)**
Security Headers Compliance	**Both Passed HSTS & CSP**

50+ Signature Tech Profiler

Detects CMS platforms, JavaScript frameworks, CDNs, and server runtimes in <100ms:

- **Frameworks:** Next.js, React, Vue, Nuxt, Angular, Svelte
- **CMS:** WordPress, Shopify, Webflow, Ghost, Drupal
- **Hosting/CDNs:** Cloudflare, Vercel, AWS CloudFront, Fastly
- **Analytics:** Google Analytics 4, Tag Manager, Hotjar

PRODUCT CAPABILITIES & MICRO-TOOLS

9 Specialized Diagnostic Engines

A unified suite engineered for technical SEO consultants, webmasters, agencies, and autonomous AI agents.

01 • AUDIT

SEO Health Scorer

0–100 deterministic scoring with radial SVG gauge.

02 • COMPETITIVE

Competitor Benchmark

Side-by-side URL compare with winner metrics.

03 • WEB VITALS

Speed Lab

LCP, CLS, INP, and TTFB via Google REST API.

04 • CODE STUDIO

Schema JSON-LD

Validates structured data & OpenGraph tags.

05 • STACK

50+ Tech Detector

Detects CMS, React, Next.js, and CDNs in <100ms.

06 • AEO AI

Brand Radar AI

Brand citation Share of Voice across LLMs.

07 • CRAWL

404 Broken Link Crawler

Scans anchor tags to identify dead links.

08 • INDEXING

Robots & Sitemap

Validates crawl directives & XML sitemaps.

09 • SECURITY

HTTP Headers & SSL

Inspects TLS cipher suites & CSP headers.

AI SYNTHESIS & REMEDIATION ENGINE

Automated Developer Remediation with Gemini AI

Transforming complex technical audit data into ready-to-copy code patches and executive briefs.

Structured Prompt Engineering

We feed validated audit findings JSON into Google Gemini with strict prompt constraints:

- **Zero Metric Invention:** AI cannot alter the deterministic 0–100 score.
- **Impact Prioritization:** Rank issues by Critical, Warning, and Advisory.
- **Executable Fixes:** Provide exact copy-paste code snippets for Next.js, HTML, or Apache/Nginx configs.

// SAMPLE GEMINI AI CODE PATCH

```
// Add to app/layout.tsx for missing Canonical & OpenGraph:
export const metadata: Metadata = {
  metadataBase: new URL('https://example.com'),
  alternates: {
    canonical: '/',
  },
  openGraph: {
    title: 'Example Brand — Intelligence',
    description: 'Verified technical diagnostics',
    images: ['/og-image.png'],
  },
};
```

✓ Ready for immediate one-click copy & paste by web developers.

CLOUD ARCHITECTURE & DATA PERSISTENCE

Google Cloud SQL PostgreSQL & Cloud Run

Immutable report persistence, relational schema design, and serverless horizontal autoscaling.

< 100ms

Crawl Latency

In-Memory Cheerio DOM

140 MB

Docker Image

Node 20 Alpine Standalone

0 Idle

Scale-to-Zero Pods

Stateless Cloud Run

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Routes Compiled

0 TypeScript Errors

Persistent Relational Architecture:

- **Immutable Audits Table:** Stores UUID primary keys with JSONB telemetry snapshots, preventing redundant scrapes.
- **Resilient Database Pool:** Cloud SQL connection pooling with automatic local memory fallback for offline testing.
- **Permanent Permalink Sharing:** Route `/report/[id]` provides instant, reproducible historical audits.

AI AGENT INTEROPERABILITY & MCP STANDARD

Model Context Protocol (FastMCP) for AI Agents

Connecting Claude Desktop, Cursor IDE, and autonomous agents directly to SEO Audit via JSON-RPC 2.0.

>_ What is MCP?

Model Context Protocol is Anthropic's open standard enabling LLMs to call local and cloud tools securely.

seoseite_audit_url: Autonomous audits & Core Web Vitals directly in IDE.

seoseite_compare_domains: Head-to-head competitor compare in chat.

seoseite_detect_tech: Live framework interrogation during coding.

seoseite_check_broken_links: Crawls dead links with markdown summaries.

// .CURSOR/MCP.JSON

```
{
  "mcpServers": {
    "seoseite": {
      "command": "npx",
      "args": ["-y", "@seoseite/mcp-server"],
      "env": {
        "SEOSEITE_API_KEY": "sk_live_token"
      }
    }
  }
}
```

Allows developers to type *"Audit linear.app with SEO Audit"* inside Cursor IDE or Claude.

COMPARATIVE ENGINEERING MATRIX

SEO Audit vs Traditional Web Crawlers

A direct systems engineering comparison highlighting efficiency, security, and algorithmic transparency.

ENGINEERING DIMENSION	SEO AUDIT (THIS PROJECT)	HEADLESS CHROME SCRAPERS	COMMERCIAL AUDIT TOOLS
Crawl Engine	Cheerio In-Memory DOM	Chromium (Puppeteer)	Proprietary Cluster
RAM Consumption	< 20 MB / process	1,500 – 2,000 MB	High Server Footprint
Parse Latency	Sub-100 ms average	4,000 – 8,000 ms	Queued / Asynchronous
Scoring Method	Deterministic 0–100 Formula	Lighthouse Only	Black-Box Proprietary
SSRF Security	Pre-Flight DNS IP Shield	Usually Unfiltered	Internal Whitelist
Agent Integration	Model Context Protocol (MCP)	None	Restricted REST Only
Compute Cost (10k Scans)	~$0.08 (Cloud Run)	~$14.50 (Chromium VM)	High Recurring Pricing

CONCLUSION & VIVA VOCE DEFENSE

Project Summary & Live Demonstration

SEO Audit establishes an honest, high-performance, and verifiable website intelligence platform.

Open & Transparent

Zero fabricated metrics. Full algorithmic clarity, verifiable formulas, and reproducible sources.

High Performance

Sub-second Cheerio DOM crawling, Next.js 16 SSR, and Google Cloud Run auto-scaling.

AI Agent Ready

Native Model Context Protocol (FastMCP) integration for modern autonomous AI workflows.

Thank You!

Ready for Live Demonstration & Viva Voce Defense

Launch Dashboard Demo | Launch Live Website Audit | Download All 14 Slides (PDF)